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Subject: MDM- Wireless & Mobile Communication

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Experiment No.:	04
Title:	To scan nearby Wi-Fi networks using a mobile application and analyze their SSID, RSSI (signal strength), operating channel, frequency band, and channel occupancy.
Date of Performance:	
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Sign of Faculty:	

Aim: To scan nearby Wi-Fi networks using a mobile application and analyze their SSID, RSSI (signal strength), operating channel, frequency band, and channel occupancy.

Objective: The objectives of this experiment are:

- To identify nearby Wi-Fi networks.
- To understand the concept of SSID.
- To measure Wi-Fi signal strength using RSSI.
- To observe the Wi-Fi channels used by different networks.
- To study channel congestion and interference.
- To compare signal strength at different locations.

Theory:

Wi-Fi Network Scanning

Wi-Fi is one of the most widely used wireless communication technologies, enabling smartphones, laptops, tablets, and Internet of Things (IoT) devices to communicate over radio-frequency signals without requiring physical cables. A Wi-Fi network is created using a wireless access point or router that continuously broadcasts radio signals within its coverage area. These signals allow nearby devices to discover available networks and establish wireless communication.

Wi-Fi network scanning is the process of detecting nearby wireless networks by listening to the beacon signals periodically transmitted by Wi-Fi access points. During the scanning process, a mobile device collects important information about each available network, such as its Service Set Identifier (SSID), signal strength, operating frequency, channel number, security type, and hardware address (BSSID). This information helps users identify available networks and assists network administrators in evaluating wireless coverage, channel utilization, and possible interference.

Unlike network penetration or security testing, Wi-Fi scanning simply observes publicly broadcast information and does not require connecting to any network. Therefore, it is a safe and non-intrusive method for analyzing wireless communication environments.

Service Set Identifier (SSID)

The Service Set Identifier (SSID) is the unique name assigned to a Wi-Fi network. It is the network name displayed when a user searches for available Wi-Fi connections on a smartphone or computer. The SSID helps users distinguish one wireless network from another and select the desired network for connection.

Examples of SSIDs include:

- VCET_WIFI
- Library_WiFi
- Home_Network
- Office_WiFi

Although multiple access points can share the same SSID to provide seamless coverage within a large campus or building, each access point has its own unique hardware address called the Basic Service Set Identifier (BSSID).

Received Signal Strength Indicator (RSSI)

The Received Signal Strength Indicator (RSSI) is a measurement of the power level of a received Wi-Fi signal. It indicates how strongly a wireless device receives signals from a nearby access point. RSSI is generally measured in decibel-milliwatts (dBm) and is represented using negative values.

A value closer to zero indicates a stronger signal.

For example:

RSSI (dBm)	Signal Quality
-30 to -50	Excellent
-51 to -60	Very Good
-61 to -70	Good

-71 to -80	Weak
Below -80	Very Weak

For example, a signal strength of -42 dBm is much stronger than -75 dBm because -42 is closer to zero.

RSSI is affected by several factors including distance from the access point, walls, furniture, obstacles, electromagnetic interference, and the number of users connected to the network. As the distance from the router increases, the RSSI value decreases, resulting in weaker signal reception.

Wi-Fi Frequency Bands

Wi-Fi communication primarily operates in three frequency bands:

Frequency Band	Applications
2.4 GHz	Home Wi-Fi, IoT devices, Bluetooth coexistence
5 GHz	High-speed Wi-Fi, HD video streaming, gaming
6 GHz	Wi-Fi 6E and Wi-Fi 7 applications

The **2.4 GHz** band provides longer communication range but is more susceptible to interference because many wireless devices operate within this frequency band.

The **5 GHz** band provides higher data rates with reduced interference but has a shorter coverage range.

The **6 GHz** band supports the latest Wi-Fi standards and offers wider bandwidth and lower congestion for high-performance wireless communication.

Wi-Fi Channels

Each Wi-Fi frequency band is divided into multiple communication channels. These channels allow several wireless networks to operate simultaneously within the same frequency band.

In the **2.4 GHz** band, the most commonly used non-overlapping channels are:

- Channel 1
- Channel 6
- Channel 11

If multiple nearby Wi-Fi networks use the same or overlapping channels, radio interference increases, resulting in reduced throughput, higher latency, and lower network performance. Therefore, selecting a less congested channel improves communication quality and network reliability.

Channel Occupancy

Channel occupancy refers to the number of Wi-Fi networks simultaneously operating on a particular wireless channel. A channel with many nearby access points is considered congested because multiple devices compete for the same wireless medium.

A Wi-Fi analyzer application graphically displays channel occupancy, enabling users to identify:

- Busy channels
- Less congested channels
- Overlapping Wi-Fi signals
- Strong and weak wireless networks

Network administrators often use channel occupancy analysis while designing wireless networks to minimize interference and maximize network performance.

Wi-Fi Analyzer Mobile Application

A Wi-Fi Analyzer is a mobile application that scans nearby wireless networks and presents detailed information regarding their operating characteristics. It continuously listens to beacon frames transmitted by nearby access points and displays parameters such as:

- SSID
- BSSID
- RSSI
- Channel Number
- Frequency Band
- Security Type
- Channel Graph
- Signal Strength Graph

These applications are widely used by network engineers, students, and technicians for wireless site surveys, network troubleshooting, channel planning, and performance optimization.

Working Principle

When the Wi-Fi Analyzer application is started, the smartphone's wireless adapter scans all nearby access points by listening for beacon signals. The application collects information transmitted in these beacon frames and displays the detected wireless networks. Students record the SSID, RSSI, operating channel, and frequency band of each detected network and analyze channel occupancy to understand wireless communication behavior in different locations.

Step-by-Step Procedure

Step 1: Install and Open the Application

Install the Wi-Fi analyzer application on your Android smartphone.

Open the application and allow the required permissions. On modern Android versions, Wi-Fi scanning may require Location permission and Location Services to be enabled because of the operating system's Wi-Fi scanning requirements. (Google Play)

Step 2: Go to an Appropriate Location

Perform the experiment in an area where multiple Wi-Fi networks are available, such as:

- College laboratory
- Classroom
- Office
- Home
- Public area

For an academic experiment, it is better to perform the scan at two or three different locations and compare the results.

For example:

- Location A: Laboratory
- Location B: Corridor
- Location C: Classroom

Step 3: Start the Wi-Fi Scan

Open the **Access Points / Network List** section of the application.

The application will display nearby Wi-Fi networks.

You may see information such as:

- SSID
- BSSID
- RSSI
- Channel
- Frequency
- Wi-Fi band
- Security type

For your experiment, you primarily need:

SSID + RSSI + Channel

Step 4: Record the SSID

Write down the name of each detected Wi-Fi network.

Example:

Sr. No.	SSID
1	College_WiFi
2	Faculty_WiFi
3	Student_WiFi

Note: Do not record passwords. You are only observing publicly broadcast Wi-Fi network information.

Step 5: Record RSSI

For each Wi-Fi network, record the signal strength.

Example:

SSID	RSSI
College_WiFi	-42 dBm
Faculty_WiFi	-58 dBm
Student_WiFi	-74 dBm

Interpretation:

- `-42 dBm` → Strong signal
- `-58 dBm` → Good signal
- `-74 dBm` → Relatively weak signal

Step 6: Record the Wi-Fi Channel

Record the channel number used by each network.

Example:

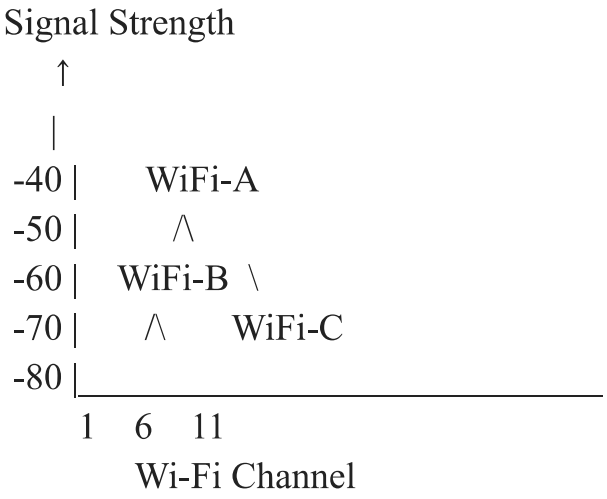
SSID	RSSI	Channel
College_WiFi	-42 dBm	1
Faculty_WiFi	-58 dBm	6
Student_WiFi	-74 dBm	11

This is the basic channel mapping.

Channel Mapping

The application normally provides a graphical view of Wi-Fi networks.

For example:



The graph allows you to see:

- Which channels are occupied

- Which networks have strong signals
- Whether several networks overlap on the same channel
- Which channel appears less crowded

The WiFi Analyzer application specifically provides channel graphs and channel analysis features for nearby access points. (Google Play)

Sample Observation Table

Refer table and create own observation table:

Sr. No	SSID	RSSI (dBm)	Channel	Frequency Band	Signal Quality
1	VCET_WIFI	-45	1	2.4 GHz	Excellent
2	Faculty_WiFi	-58	6	2.4 GHz	Very Good
3	Library_WiFi	-67	11	2.4 GHz	Good
4	Office_WiFi	-76	6	2.4 GHz	Weak

Observation Table

Fill the values in below observation Table:

Sr. No.	SSID	RSSI (dBm)	Channel No.	Frequency Band	Signal Quality
1	EXAMSECTION	-66	1	2.4GHz	Excellent
2	VCET-5G	-74	36(42)	5GHz	Poor
3	GSVCET	-58	149(155)	5GHz	Excellent
4	office_conference	-73	11	2.4GHz	Poor

Students should record their actual observations.

Students Should Analyze:

After collecting the data, ask students to answer the following:

Question 1

Which Wi-Fi network has the strongest signal?

Answer: The network with the RSSI value closest to zero.

For example:

- WiFi_A = -45 dBm
- WiFi_B = -70 dBm

Therefore, WiFi_A has the stronger signal.

Question 2

Which channel has the highest number of networks?

Count the number of networks using each channel.

Example:

Channel	Number of Networks
1	3
6	5
11	2

In this example, Channel 6 is the most crowded.

Question 3

Which channel appears least crowded?

The channel having fewer nearby networks is considered less congested.

Question 4

Is the strongest signal always the best network?

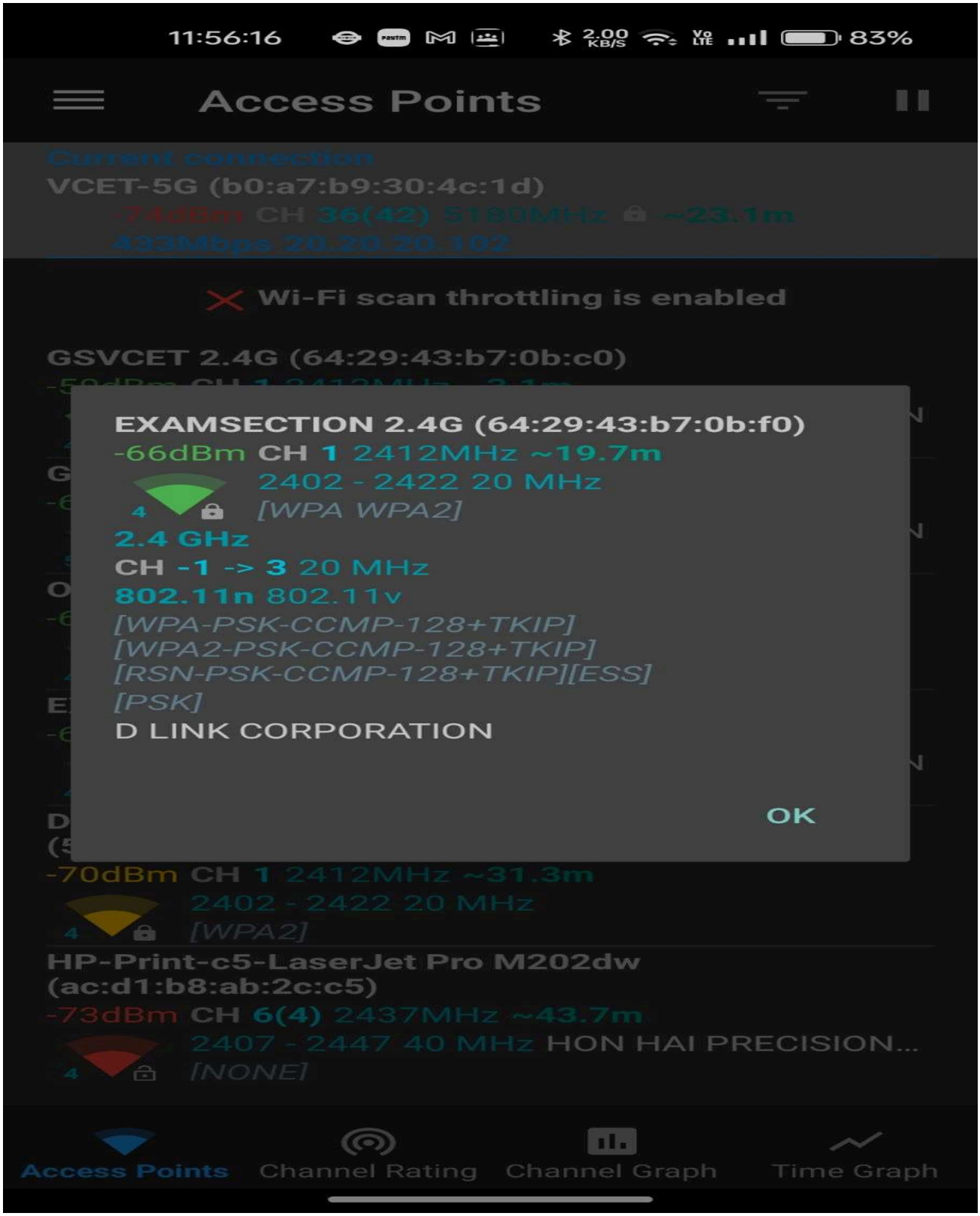
Not necessarily.

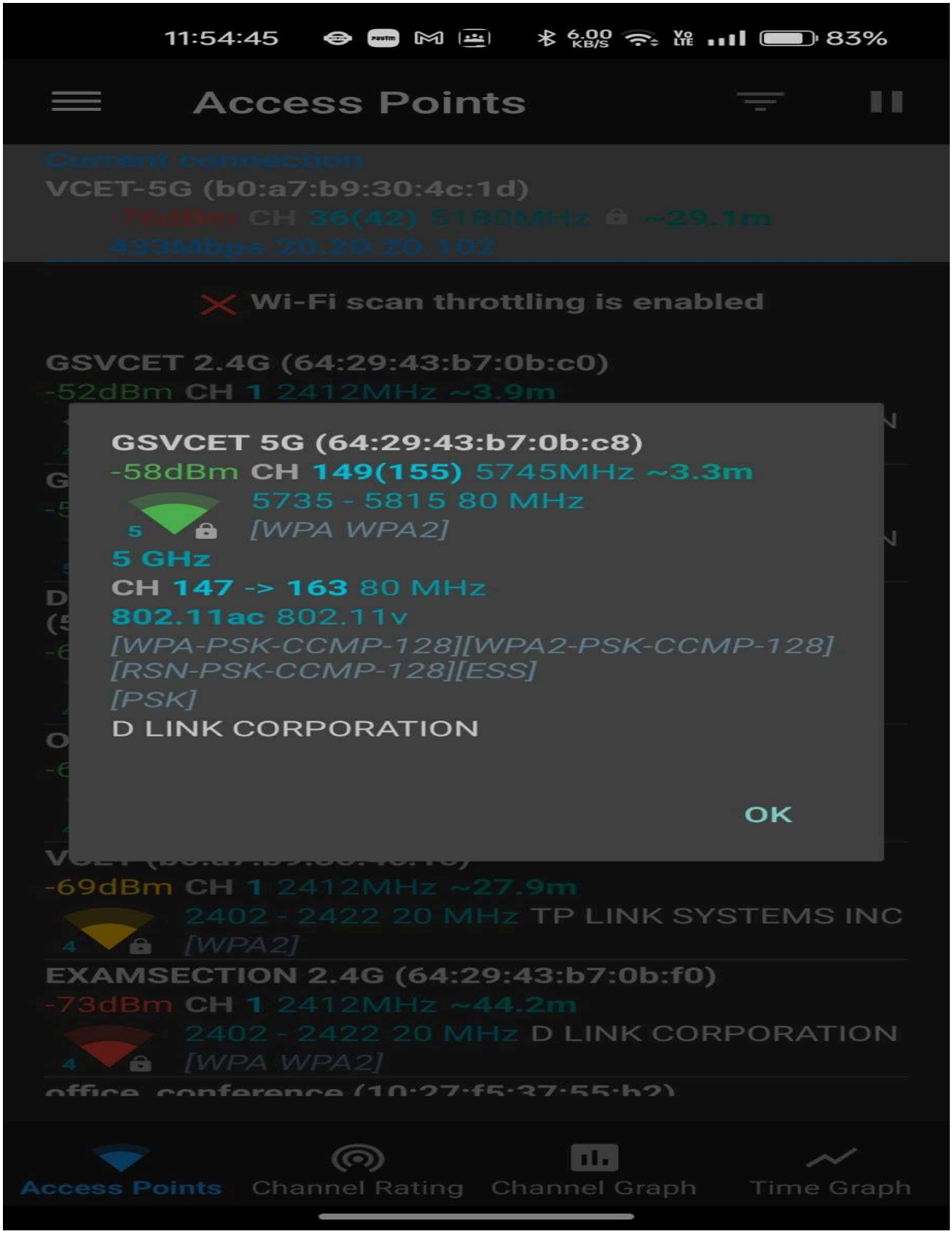
A strong signal is important, but network performance also depends on:

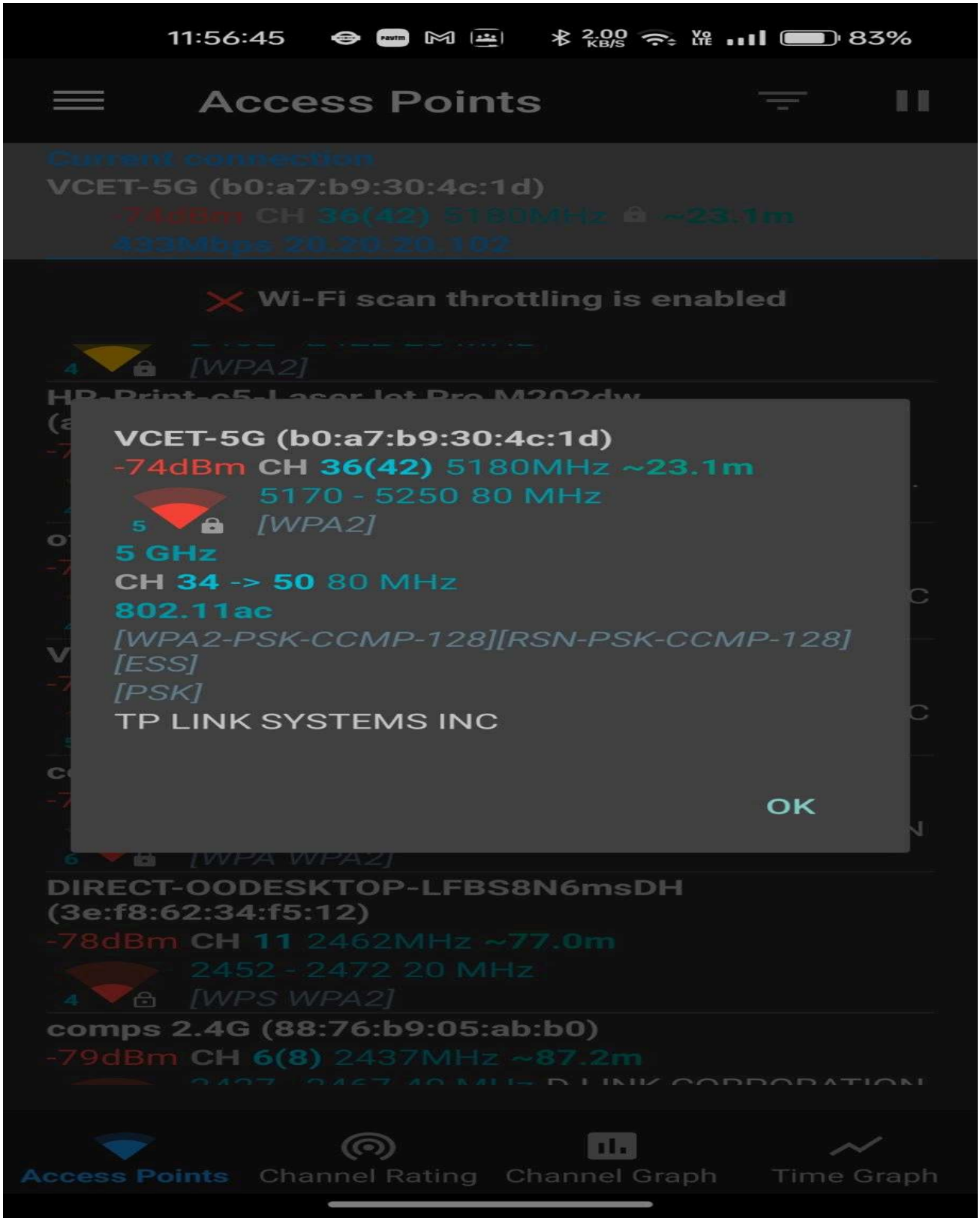
- Channel congestion
- Interference
- Network bandwidth
- Number of connected users
- Internet connection speed

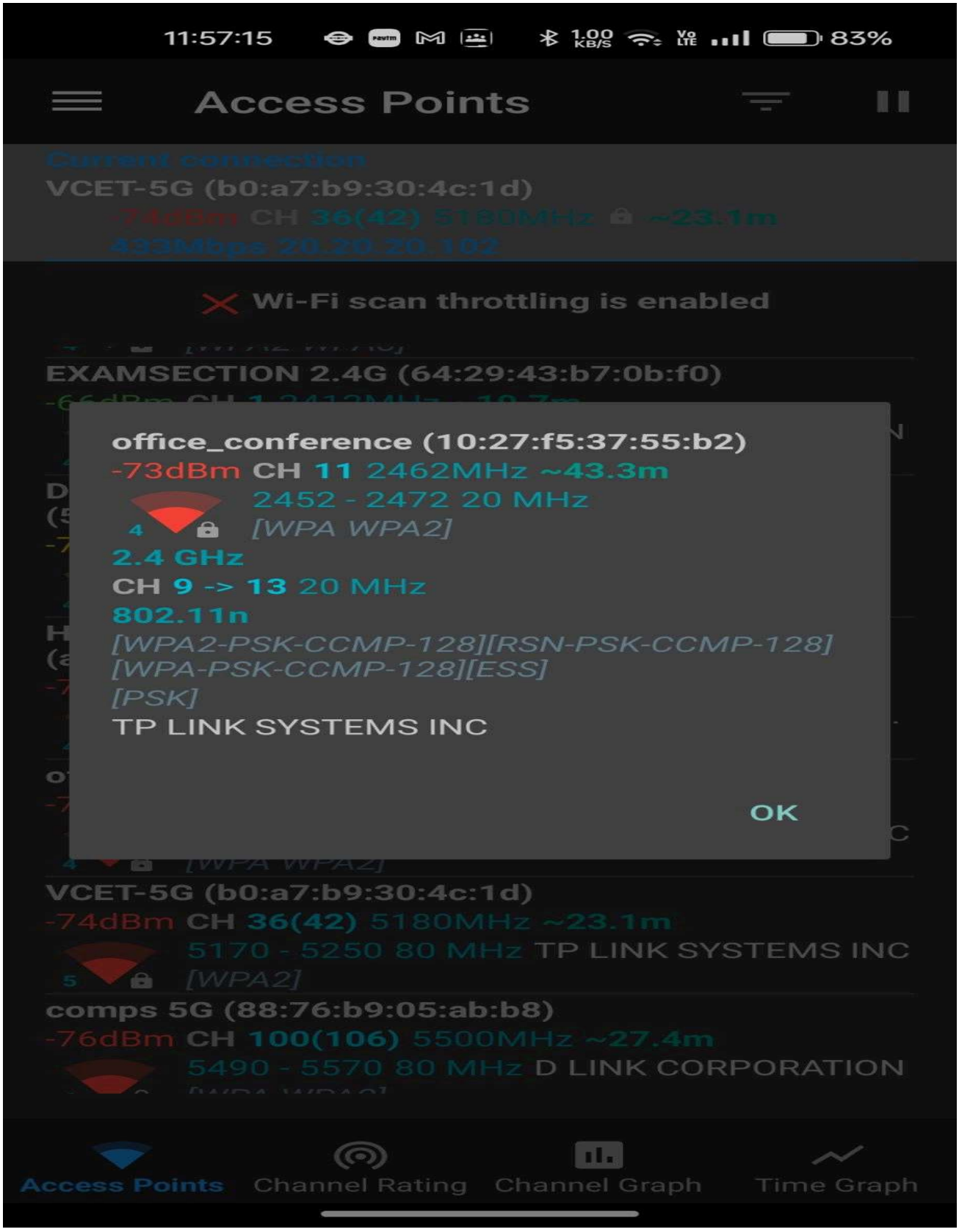
Code/Output:


11:41:26
85%
Access Points
Current connection
VCET-5G (b0:a7:b9:30:4c:1d)
-71dBm CH 36(42) 5180MHz ~16.3m
433Mbps 20.20.20.102
Wi-Fi scan throttling is enabled
OnePlus Nord CE4 (6a:d9:bb:ec:b6:cc)
-53dBm CH 11 2462MHz ~4.3m
2452 - 2472 20 MHz
[WPA2 WPA3]
GSVCET 2.4G (64:29:43:b7:0b:c0)
-55dBm CH 1 2412MHz ~5.6m
2402 - 2422 20 MHz D LINK CORPORATION
[WPA WPA2]
GSVCET 5G (64:29:43:b7:0b:c8)
-58dBm CH 149(155) 5745MHz ~3.3m
5735 - 5815 80 MHz D LINK CORPORATION
[WPA WPA2]
Rugved (26:7e:2b:ea:9c:94)
-59dBm CH 149(155) 5745MHz ~3.7m
5735 - 5815 80 MHz
[WPA2]
office_conference (10:27:f5:37:55:b2)
-67dBm CH 11 2462MHz ~21.7m
2452 - 2472 20 MHz TP LINK SYSTEMS INC
[WPA WPA2]
VCET-5G (b0:a7:b9:30:4c:1d)
-71dBm CH 36(42) 5180MHz ~16.3m
5170 - 5250 80 MHz TP LINK SYSTEMS INC
[WPA2]
HP-Print-c5-LaserJet Pro M202dw
Access Points
Channel Rating
Channel Graph
Time Graph


11:56:16
83%
Access Points
Current connection
VCET-5G (b0:a7:b9:30:4c:1d)
-74dBm CH 36(42) 5180MHz ~23.1m
433Mbps 20.20.20.102
Wi-Fi scan throttling is enabled
GSVCET 2.4G (64:29:43:b7:0b:c0)
EXAMSECTION 2.4G (64:29:43:b7:0b:f0)
-66dBm CH 1 2412MHz ~19.7m
2402 - 2422 20 MHz
[WPA WPA2]
2.4 GHz
CH -1 -> 3 20 MHz
802.11n 802.11v
[WPA-PSK-CCMP-128+TKIP]
[WPA2-PSK-CCMP-128+TKIP]
[RSN-PSK-CCMP-128+TKIP][ESS]
[PSK]
D LINK CORPORATION
OK
-70dBm CH 1 2412MHz ~31.3m
2402 - 2422 20 MHz
[WPA2]
HP-Print-c5-LaserJet Pro M202dw
(ac:d1:b8:ab:2c:c5)
-73dBm CH 6(4) 2437MHz ~43.7m
2407 - 2447 40 MHz HON HAI PRECISION...
[NONE]
Access Points
Channel Rating
Channel Graph
Time Graph


11:54:45
83%
Access Points
Current connection
VCET-5G (b0:a7:b9:30:4c:1d)
-76dBm CH 36(42) 5180MHz ~29.1m
433Mbps 20.20.20.102
Wi-Fi scan throttling is enabled
GSVCET 2.4G (64:29:43:b7:0b:c0)
-52dBm CH 1 2412MHz ~3.9m
GSVCET 5G (64:29:43:b7:0b:c8)
-58dBm CH 149(155) 5745MHz ~3.3m
5735 - 5815 80 MHz
[WPA WPA2]
5 GHz
CH 147 -> 163 80 MHz
802.11ac 802.11v
[WPA-PSK-CCMP-128][WPA2-PSK-CCMP-128]
[RSN-PSK-CCMP-128][ESS]
[PSK]
D LINK CORPORATION
OK
-69dBm CH 1 2412MHz ~27.9m
2402 - 2422 20 MHz TP LINK SYSTEMS INC
[WPA2]
EXAMSECTION 2.4G (64:29:43:b7:0b:f0)
-73dBm CH 1 2412MHz ~44.2m
2402 - 2422 20 MHz D LINK CORPORATION
[WPA WPA2]
Access Points
Channel Rating
Channel Graph
Time Graph


11:56:45
83%
Access Points
Current connection
VCET-5G (b0:a7:b9:30:4c:1d)
-74dBm CH 36(42) 5180MHz ~23.1m
433Mbps 20.20.20.102
Wi-Fi scan throttling is enabled
[WPA2]
VCET-5G (b0:a7:b9:30:4c:1d)
-74dBm CH 36(42) 5180MHz ~23.1m
5170 - 5250 80 MHz
[WPA2]
5 GHz
CH 34 -> 50 80 MHz
802.11ac
[WPA2-PSK-CCMP-128][RSN-PSK-CCMP-128]
[ESS]
[PSK]
TP LINK SYSTEMS INC
OK
DIRECT-OODESKTOP-LFBS8N6msDH
(3e:f8:62:34:f5:12)
-78dBm CH 11 2462MHz ~77.0m
2452 - 2472 20 MHz
[WPS WPA2]
comps 2.4G (88:76:b9:05:ab:b0)
-79dBm CH 6(8) 2437MHz ~87.2m
Access Points
Channel Rating
Channel Graph
Time Graph


11:57:15
83%
Access Points
Current connection
VCET-5G (b0:a7:b9:30:4c:1d)
-74dBm CH 36(42) 5180MHz ~23.1m
433Mbps 20.20.20.102
Wi-Fi scan throttling is enabled
EXAMSECTION 2.4G (64:29:43:b7:0b:f0)
office_conference (10:27:f5:37:55:b2)
-73dBm CH 11 2462MHz ~43.3m
2452 - 2472 20 MHz
[WPA WPA2]
2.4 GHz
CH 9 -> 13 20 MHz
802.11n
[WPA2-PSK-CCMP-128][RSN-PSK-CCMP-128]
[WPA-PSK-CCMP-128][ESS]
[PSK]
TP LINK SYSTEMS INC
OK
VCET-5G (b0:a7:b9:30:4c:1d)
-74dBm CH 36(42) 5180MHz ~23.1m
5170 - 5250 80 MHz TP LINK SYSTEMS INC
[WPA2]
comps 5G (88:76:b9:05:ab:b8)
-76dBm CH 100(106) 5500MHz ~27.4m
5490 - 5570 80 MHz D LINK CORPORATION
Access Points
Channel Rating
Channel Graph
Time Graph

Conclusion

The nearby Wi-Fi networks were successfully scanned using a mobile Wi-Fi analyzer application. The SSID, RSSI, operating channels, frequency bands, and channel occupancy of the detected wireless networks were recorded and analyzed. The experiment demonstrated that RSSI indicates signal strength, while channel occupancy helps identify congested and less congested channels. The practical study provided a clear understanding of how Wi-Fi networks operate and how proper channel selection contributes to efficient and reliable wireless communication.